Python Engineering - Week 02

Week 02 - Data structures

`01_python_bootcamp/week_02_data_structures/`

How to use this document

This PDF is a review index for the week. Each class summarizes **the problem, concepts and syntax/functions introduced**. If a concept becomes fuzzy, locate it here and return to the original exercise.

class_05_lists_tuples

EX13 - Trajectory data with lists and tuples

List: Ordered mutable collection: elements can be changed, appended or removed.

Tuple: Ordered structure whose container layout does not change; useful for snapshots and multiple returns.

Syntax: my_list = [...], append(...), indexing [i], slicing [a:b], my_tuple = (...).

Concepts: mutability, aliasing/references and state snapshots (h, v, t).

Caution: A tuple may contain mutable objects; placing a list inside does not make the list immutable.

Key idea: Choose the structure based on whether data must grow/change or represent a compact stable state.

class_06_dictionaries_sets

EX14 - Configuration and required parameters

Dictionary: Key -> value mapping; useful for configuration and labelled data.

Set: Collection of unique elements optimized for membership and set operations.

Syntax: {'g': 9.81}, config['g'], key in config, required - provided.

Concepts: keys, values, membership, required keys and set differences.

Application: Validate that a configuration contains all required parameters.

Key idea: dict maps meaning to a value; set is excellent for presence and set comparison.

class_07_numpy_arrays

EX15 + EX15b - NumPy fundamentals

Array: Homogeneous numerical structure designed for vectorized computation.

Properties: shape, ndim, size and dtype.

Selection: indexing, slicing, rows and columns.

Operations: broadcasting, reshape, transpose, np.vstack, np.hstack, max and argmax.

Algebra: * = element-wise multiplication; @ = matrix multiplication.

Axes: axis=0 reduces rows -> result per column; axis=1 reduces columns -> result per row.

Application: Numerical trajectories and vectorized energy.

class_08_data_processing

EX16 + EX16b - Masks, filtering and analysis

Masks: Boolean arrays used to select data satisfying conditions.

Syntax: &, |, ~ to combine NumPy conditions; A[mask] to filter.

Functions: np.sum(mask), np.where(mask), np.diff, np.allclose and np.column_stack.

Analysis: basic statistics, indices, sample-to-sample changes and numerical checks.

Caution: ~(v > 0) means v <= 0; it does not necessarily mean strictly descending.

Key idea: Process datasets by selecting subsets and applying operations along specific axes.

Quick concept index

Concept	Where to find it
List: Ordered mutable collection: elements can be changed, appended or removed.	class_05_lists_tuples
Tuple: Ordered structure whose container layout does not change; useful for snapshots and multiple returns.	class_05_lists_tuples
Syntax: my_list = [...], append(...), indexing [i], slicing [a:b], my_tuple = (...).	class_05_lists_tuples
Concepts: mutability, aliasing/references and state snapshots (h, v, t).	class_05_lists_tuples
Caution: A tuple may contain mutable objects; placing a list inside does not make the list immutable.	class_05_lists_tuples
Key idea: Choose the structure based on whether data must grow/change or represent a compact stable state.	class_05_lists_tuples
Dictionary: Key -> value mapping; useful for configuration and labelled data.	class_06_dictionaries_sets
Set: Collection of unique elements optimized for membership and set operations.	class_06_dictionaries_sets
Syntax: {'g': 9.81}, config['g'], key in config, required - provided.	class_06_dictionaries_sets
Concepts: keys, values, membership, required keys and set differences.	class_06_dictionaries_sets
Application: Validate that a configuration contains all required parameters.	class_06_dictionaries_sets
Key idea: dict maps meaning to a value; set is excellent for presence and set comparison.	class_06_dictionaries_sets
Array: Homogeneous numerical structure designed for vectorized computation.	class_07_numpy_arrays
Properties: shape, ndim, size and dtype.	class_07_numpy_arrays
Selection: indexing, slicing, rows and columns.	class_07_numpy_arrays
Operations: broadcasting, reshape, transpose, np.vstack, np.hstack, max and argmax.	class_07_numpy_arrays
Algebra: * = element-wise multiplication; @ = matrix multiplication.	class_07_numpy_arrays
Axes: axis=0 reduces rows -> result per column; axis=1 reduces columns -> result per row.	class_07_numpy_arrays
Application: Numerical trajectories and vectorized energy.	class_07_numpy_arrays

Review rule

First locate **the concept**; then open that class PDF; finally run or modify the exercise. Document + code is the review mechanism for this course.